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DDOS Attack Aware DSR Routing Protocol in WSN

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Abstract

Open nature of wireless sensor networks (WSN) makes it vulnerable to outside attacks. Many security threats like denial of service, black hole, sinkhole etc. may affect the network performance. Distributed denials of service (DDOS) attacks are defined as attacks that are launched by a set of malicious entities towards a node or set of nodes. In this work we propose a solution to prevent WSN from DDOS attack using dynamic source routing (DSR). Energy of concerned nodes has been used for detection and prevention of attack. Qualnet 5.2 simulator is used for implementation of the proposed solution.

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*Keywords:*Wireless Sensor Networks; DDOS attack; DSR protocol; Battery charge

1. Introduction

A powerful combination of distributed sensing, computing and communication is achieved in Wireless Sensor networks (WSN). These networks have countless applications and, at the same time, offer several challenges[1]. WSNs help to set up various real world applications like military, surveillance, investigation more effective and dynamic. Sensor network applications help to detect instantaneous sensed values and proceed to achieve desiredtask. Small data storage capacity, low bandwidth, low power battery and low computational power and open nature make it more complex and vulnerable to many security threats[2,3].

Security threats are classified in two categories passive and active attacks. In passive attack attacker can listen the packets in the network while in the active attack attacker can also modify the packet contents. Subsequently, little attack may lie into both categories[3,4]. DDOS attack is an attempt to make network resources unavailable and

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complex task for users. It may achieve by deadlock creation among network resources or wastage of resource capabilities i.e. battery capacity, bandwidth etc. It doesn't cause any variation in message intend but overwhelm the network traffic[5]. Apart from intended destination, other nodes of the route also suffer from attack, may be in terms of draining off battery[6]. The major issue with routing protocols is that, these are designed for better performance of the network not for security of node. Secure protocols are generally designed to have features such as authentication, integrity, confidentiality and non-repudiation. For security purpose DSR have vulnerabilities and it is easily manipulated by malicious node to destroy its network routing[7,8]. Therefore the objective of this study is to make DSR protocol DDOS attack aware. This solution detects and prevents the DDOS attack using battery charge of affected nodes in WSN to avoid any bad circumstance.

2. Literature Review

Authors in[1] presented a survey of sensor network addressing the impact of various factors such as cost, environment effect etc on the network performance. A mechanism to overcome the genuine node from unwanted loss is proposed in[2], considering energy as one of the leading challenges in the network. Authors in[3] studied various design constraints in WSN i.e. power, memory & processing with exploring various threats. Dynamic and intelligent security approaches to enhance protection level and reduce security overhead are developed. An advanced DDOS attack, Service Discovery Protocol based attack, is proposed in[5] to make heavy power drain on node. This attack effectively make it inoperable and reducing the battery life by as much as 97%. Moreover, it sends new service request similar to SYN Flooding to create conjunction and overwhelming the nodes. DDOS attack by SYN flooding is implemented in[6] on ad-hoc on-demand distance vector routing (AODV) routing protocol. Performance is evaluated by considering impact of service request and benign power attacks and studied power drain. To prevent low level DDOS attack, a scheme is presented in[7] for home based wireless sensor networks. Filtering of undesirable packets has also been presented in this paper. In[8] authors observed that vampire attack is one of severe attack which may be serious for power drain. Vampire attack is the enhance version of DDOS attack which suddenly reason for power drain. Since Vampires use protocol-compliant messages, these attacks are very difficult to detect and prevent. In[9] authors proposed a mechanism to detect and prevent genuine nodes from vampire attack.

Different routing protocols considering different security attacks have been proposed in the literature[10], such as AODV has been modified against Black-hole attack by including new Packet Route Checker. Very few of these protocols are considering the battery status of nodes while evaluating the performance of security aware routing protocol. In this paper we have applied the security mechanism in DSR routing protocol, to protect it from DDOS attack. Work has been carried out in four steps as: (i) Inclusion of battery and energy models in required source code files, which will facilitate energy measurement of each node, (ii) Performance evaluation of a WSN in terms of energy using DSR protocol without employing DDOS attack, (iii) Performance evaluation of a WSN using DSR protocol with employing DDOS attack, (iv) Prevention from DDOS attack.

3. Related Theory

3.1. Routing Protocols

Routing protocols are used to discover routes first, then responsible for transmission of packets and repairing of routes. It is the functionality of the network layer of the network[7]. There are three types of routing protocols: Proactive, Reactive and Hybrid. Proactive protocols maintain tables and driven by it. These protocols monitor the network topology continuously and evaluate the routes instantly. Optimized link state routing (OLSR) and destination sequenced distance vector (DSDV) are the examples of these protocols. Reactive protocols on the other hand, are based on demand and discover the route initially before the transmission. Once a route is established, it is maintained in the routing table until the destination is out of reach or the route expires. DSR is the kind of reactive protocol. Alternatively, hybrid routing protocols combines the merits of both proactive and reactive routing protocols.

3.2. DDOS Attack

DDOS is a type of DOS attack, where multiple compromised systems which are usually infected with a Trojan are used to target a single system causing a Denial of Service (DOS) attack[4]. It attempts to reduce or zero-out the operational capabilities of the victim from one or multiple locations. The victim of such an attack can either be a single node, a set of nodes, the base station, or even the entire network. These attacks can further classified as flooding, ping of death, Smurf attack and flooding on victim's link. There are different ways in which DDOS attack can be implemented. These are targeted flooding technique, moveable node congestion, one owner and slave congestion etc.

4. Proposed Solution

The proposed work provides a modified DSR with security aware mechanism for DDOS attack. It is carried out in four steps.

- Inclusion of battery and energy models in required source code files to facilitate energy measurement of each node -
  A qualnet 5.2 simulator is used to develop and observe the performance of proposed sensor network scenario. The simulator in its basic configuration does not permit to have the energy of each node in output window. In order to measure the energy of each node and to print the energy of each node, we require battery and energy models to be included in required source code file. A battery model "Liner battery" has been configured with known capacity at initial stage. Furthermore, "Generic" Energy model is configured to specify energy consumption in transmission, receiving, idle and sleeping stages. battery_model.cpp and battery_model.h files have been included with routing_dsr.cpp and routing_dsr.h files, to integrate battery observation on each node to keep track about battery initialization and consumption during communication. Following methods: Battery_Init, Battery_Finalize, Battery_Get_Remaining_Charge, BATTERY_RunTime_Stat, Battery_Dec_Charge are called into routing_dsr.cpp file to configure battery model with DSR routing protocol. These methods keep track on battery consumption, introducing overload during attack and calculating natural and intentional power consumption.
- Performance evaluation of a WSN in terms of energy using DSR protocol without employing DDOS attack.
  In this step a WSN is setup with homogeneous nodes using DSR routing protocol keeping specific initial energy of each node. After running the scenario we get output energy of nodes due to DSR mechanism only. No DDOS attack has been included in this phase.
- Performance evaluation of a WSN using DSR protocol with employing DDOS attack
  This phase includes detection of DDOS attack. In this work DDOS attack has been implemented using targeted flooding. Deployment of multiple malicious nodes is done with more battery capacity than non malicious nodes. These malicious nodes decrease the battery power of any node coming into route of transmission. By this method victim node's power goes down to very low value.
- Prevention of DDOS attack
  Examining battery charge of each node provides identification of malicious node. Because DSR doesn't have any Blacklist for sensor network, a shutdown method has been used to ignore malicious node in the network. It will step out malicious node from communication and start transferring packet transmission from alternative routes. Finally, Energy consumption is measured to compare performance, before and after prevention technique. Fig. 1 shows the flow diagram of proposed solution.

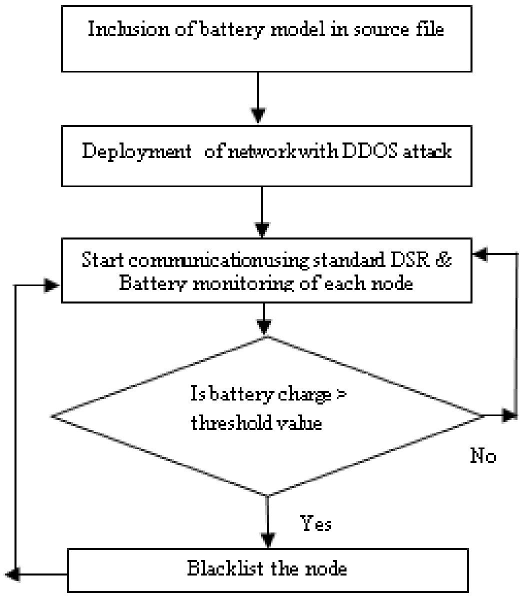

Fig. 1. Flow diagram of proposed solution

5. Results and Analysis

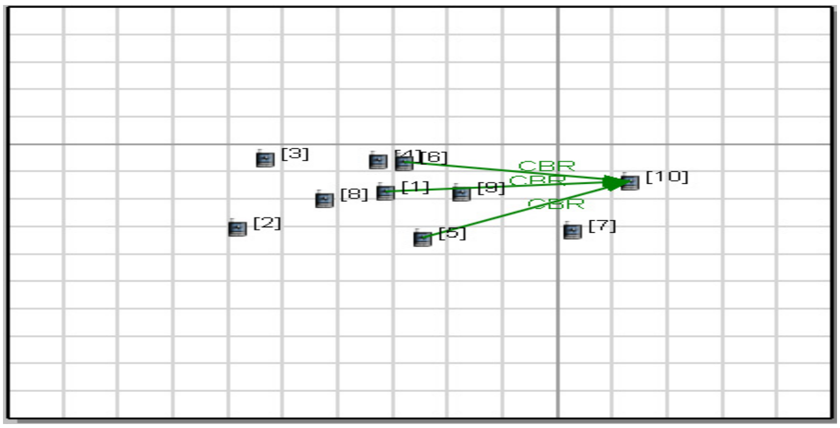

Fig. 2. Network scenario

Fig. 2 shows the initial network scenario without DDOS attack employing DSR routing protocol in WSN. Radio type is taken as 802.15.4 because we use WSN. We take 1200 mAHr battery charge initially for each node. A CBR link is connected between node 1 to node 10 and other nodes are the intermediate nodes. Initially the source node finds a route for the destination node so it broadcasts the route request packet to neighbor nodes and neighbor nodes send request to other nearby neighbor nodes and nearby neighbor nodes continuously carried on until the destination node. After reaching the destination node there are many routes which are stored in cache. Route caches contain these routes and make priority of route with minimum hop count. The route (1- 9-10), is considered which has minimum number of hops in this scenario. Initial and final energy of each node, before and after data transmission using DSR in WSN is measured, which is shown in Fig. 3. DDOS attack has not been implemented in this scenario. It is also observed that nodes, which are not in route of source and destination, also consume battery charge in basic operation of DSR.

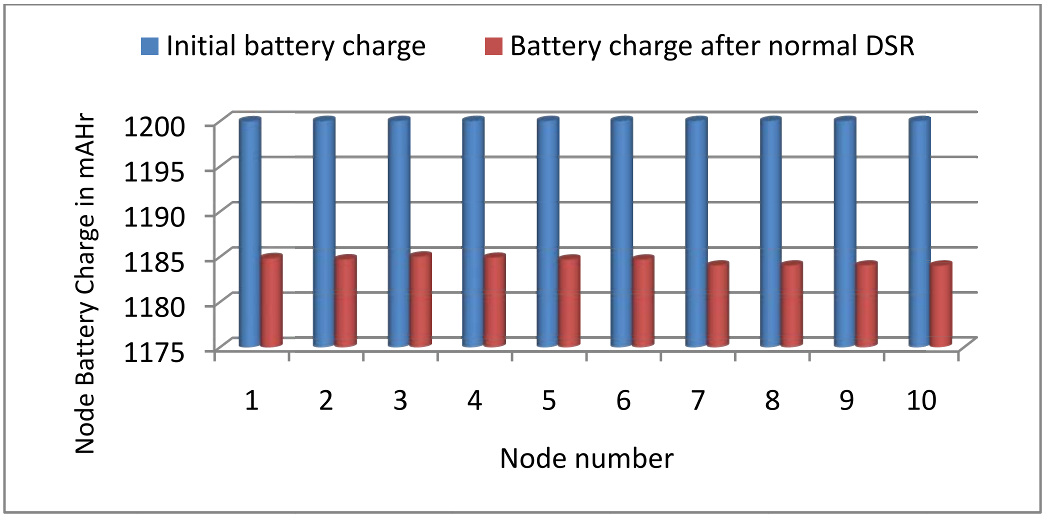

Fig. 3. Battery charge on each node without DDOS attack

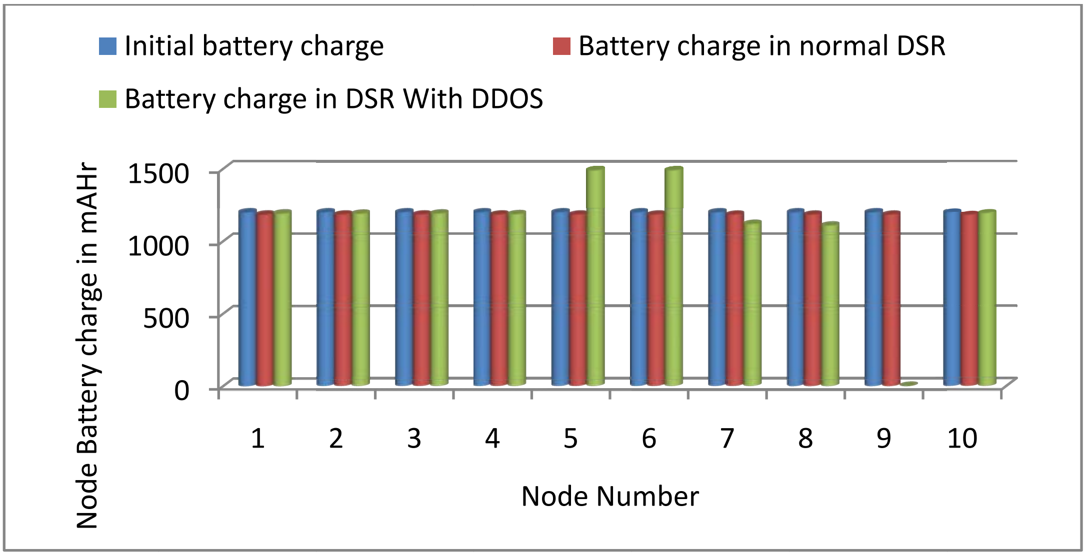

Fig. 4. Battery charge distribution of each node in presence of DDOS attack

In order to employ DDOS attack, Fig. 2 is considered again and nodes (5 & 6) are considered as malicious nodes having higher initial battery as 1500 mAhr. The CBR traffic generated is from node-1 to node-10. Other nodes are the intermediate nodes in between the source node and destination node. Malicious nodes are flooding data packets to the destination node but node 9 in between, receives these packets first, resulting in draining its energy level, which is shown in Fig. 4. When any node having more battery capacity than upper threshold value (1200 mAhr) is identified, detection method is provoked. This results in alarm by printing malicious node Id on consol. Now we get node id of malicious nodes. Shut-down method has been used to blacklist the malicious nodes. Proposed protocol blacklists malicious nodes from communication and starts transferring packet transmission from alternative routes.

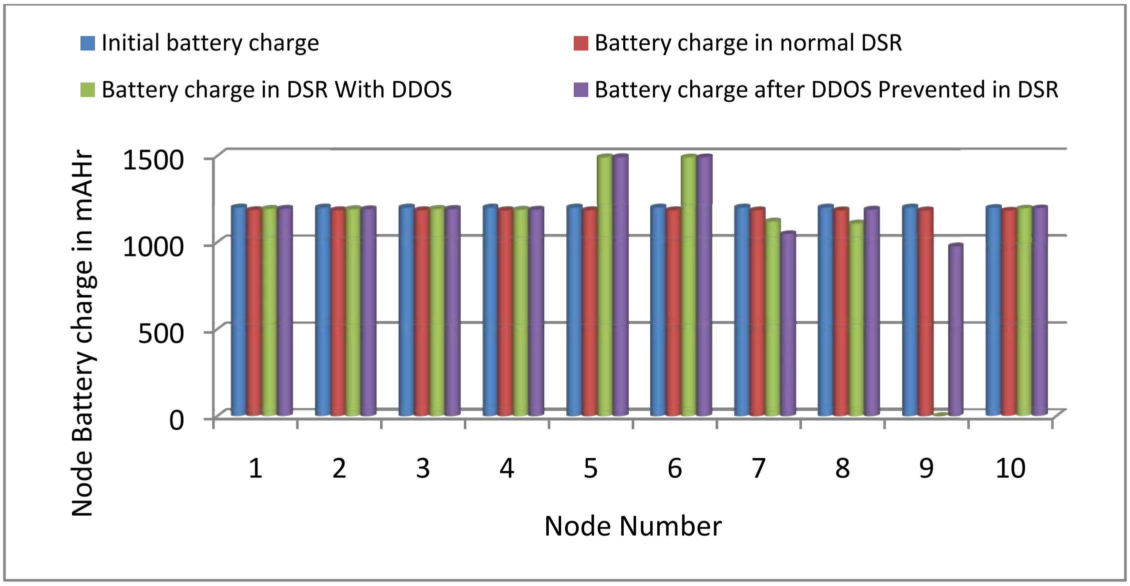

Fig. 5. Battery charge distribution of each node after prevention of DDOS attack

Finally, Energy consumption has been measured to compare energy performance of nodes before and after prevention technique in Fig. 5. It is clearly seen that at node 9, the energy level again raises. It is observed that in simple scenario energy of node 9 is 1184 mAhr, which is degraded to very low due to the DDOS attack. After employing detection and prevention method, the energy of node 9 increase to 976.29 mAhr. Since node 5 & 6 are detected as malicious nodes, data transmission will take place excluding paths in which nodes 5 & 6 are present.

6. Conclusion

WSN is vulnerable to security attacks. DDOS attack is one of severe attacks in WSN. Provisions for security mechanism in routing protocols are essential in WSN. Proposed solution considers the battery power draining for prevention and detection of DDOS attacks in DSR routing protocol, applied in WSN. Malicious nodes are easily identified and routing protocols become security aware. Proposed solution also enables longer network life because malicious nodes are shutdown and now they do not further degrade the battery charge of other nodes of the network.

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